

Assignments CITS5503 2021

Week 1

[1.0] The evolution of Cloud Computing has been compared to the evolution of electricity supply as a utility. Describe 3 specific problems that Cloud Computing solves as compared to businesses running their own data centres.

[2.0] An established financial company is about to launch their new banking application. Give 3 reasons why the company should use their own data centre rather than cloud computing.

[3.0] Describe the concepts of vertical and horizontal scale. Describe 2 different ways in which you could scale a web application horizontally. Describe a potential architecture to scale the database to handle the scaling out of the web servers.

Week 2

[4.0] Describe the different categories of services (XaaS) cloud computing can provide with specific examples of each service.

[5.0] Describe the steps which you would take on AWS and the decisions that would need to be made to create, configure and run a Virtual Machine Instance.

[6.0] Describe EBS and what features it offers

Week 3

[7.0] Describe what virtualisation is and describe the characteristic attributes of the different types of virtualisation (Language, Operating System and Hardware).

[8.0] Describe what containers are with reference to Docker and discuss their similarities and differences from operating system virtualisation perspective as provided by VirtualBox.

Week 4

[9.0] You are asked to store data about music albums in a DynamoDB table. For each album, you need to record the title of the album and the artist name. Describe the commands you would use to create a table to store such information and write an entry to that table in DynamoDB.

[10.0] Describe how S3 handles consistency of objects and how this approach affects the state of objects when they are read using a GET.

[11.0] What are the core components of DynamoDB

[12.0] When a Bucket is created, AWS allows the specification of a number of features that can be managed. What are the key properties and features?

Week 5

[13] An organisation has 5 departments and has separated out each of the IAM users into separate groups using paths following the pattern `companybucket/department1/`, `companybucket /department2/`, `companybucket /department3/*` etc.

Their IAM account names follow the pattern `user@department1.company.com`, `user@department2.company.com` etc.

You are tasked with securing a bucket that contains a folder for each of 5 departments in an organisation. Only people within a department can write to their own folder. Everyone can read from all folders.

Discuss the principles that you would use to create a policy that would achieve this objective.

Write the policy as a JSON file that you would use.

Note: you can have individual statements for each department.

[14] What aspects of security does the OSI Security Architecture X.800 standard cover? Which particular components of this standard does AWS Identity and Access Management deal with?

[15] Name 3 of the keys that you would find in a Policy. Explain their role. An example of a key is “Version” that specifies the version of the policy syntax and is normally “Version”: “2012-10-17”

Week 8

[16] Discuss the reasons why you would use Application Load Balancing and how this would be set up to load balance a Python Django application. Specifically, describe the configuration of the Listener and Target Group running the Python Django application.

[17] Describe 2 ways in which security is implemented in AWS networks at the network level. What are the similarities and differences between these 2 security implementations?

Week 9

[18] When an EC2 instance is created in AWS, it is assigned to a region and a Virtual Private Cloud (VPC). Describe how network addresses are allocated to a VPC and sub-networked when an EC2 instance is created. How is the EC2 connected to other machines and to the Internet?

[19] What is DevOps and describe how you would implement the automation of creation of machines, configuration of software and deployment of application programs using AWS.

Week 10

[20] Describe the 3 different types of Machine Learning Models that AWS Machine Learning allows you to use and describe examples of the types of questions you could answer with each one.

[21] Describe the different ways in which a Microservice Architecture can be implemented on AWS and what the benefits of this approach might be. How could authentication and authorisation be handled in this approach?

Week 11

[22] You have been asked to create a project plan for the new machine learning model your company has asked you to build. List the main tasks and sub-tasks you would need to complete to create the model on AWS. How would you measure the accuracy of the model you created?